

ESG & Financial Performance Dataset

Data analysis using Excel and SQL. Dataset from Kaggle containing simulated company data

Project Objective

Question: Is there a relationship between ESG performance and financial performance?

The aim of this project was to investigate whether companies with higher ESG scores also tend to have higher profit margins, while also exploring differences in ESG performance between industries

Dataset and Methodology

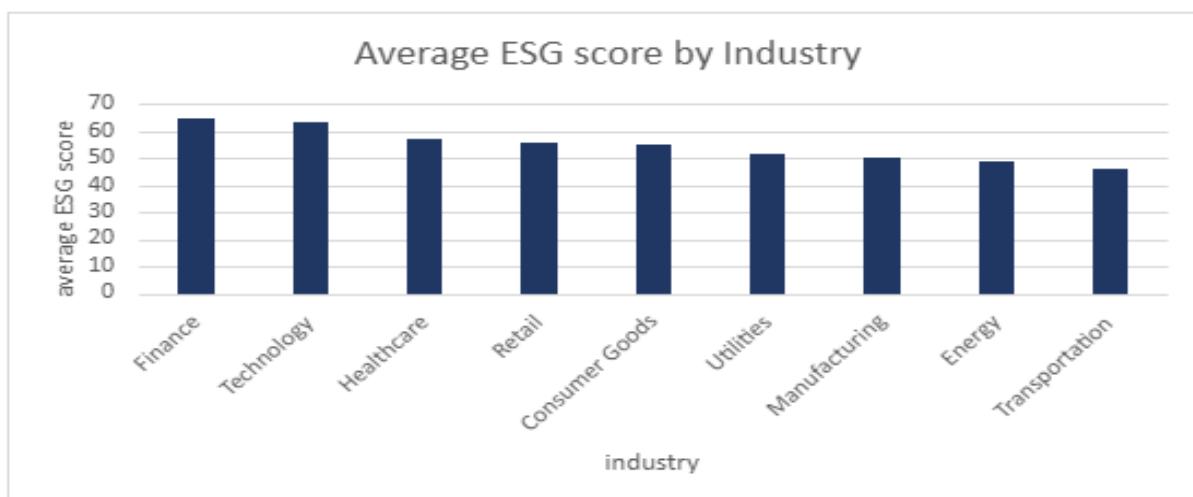
I obtained a dataset from Kaggle called the Company ESG Financial Dataset. The dataset contains information on companies across different industries and includes variables such as Company Name, Industry, Revenue, Profit Margin, Overall ESG score, Environmental ESG, Social ESG, Governance ESG and Carbon Emissions.

I used Excel to explore the dataset, create pivot tables, and produce visualisations before using SQL to analyse the data. SQL queries were used to investigate whether companies with higher ESG scores tend to have higher profit margins, which industries had the strongest ESG performance, which industries had the highest carbon emissions, and whether company size appeared to be related to ESG performance.

I used SQL functions and clauses, including SELECT, AVG (), GROUP BY, ORDER BY and CORR () to aggregate the data, compare groups, and investigate relationships between variables.

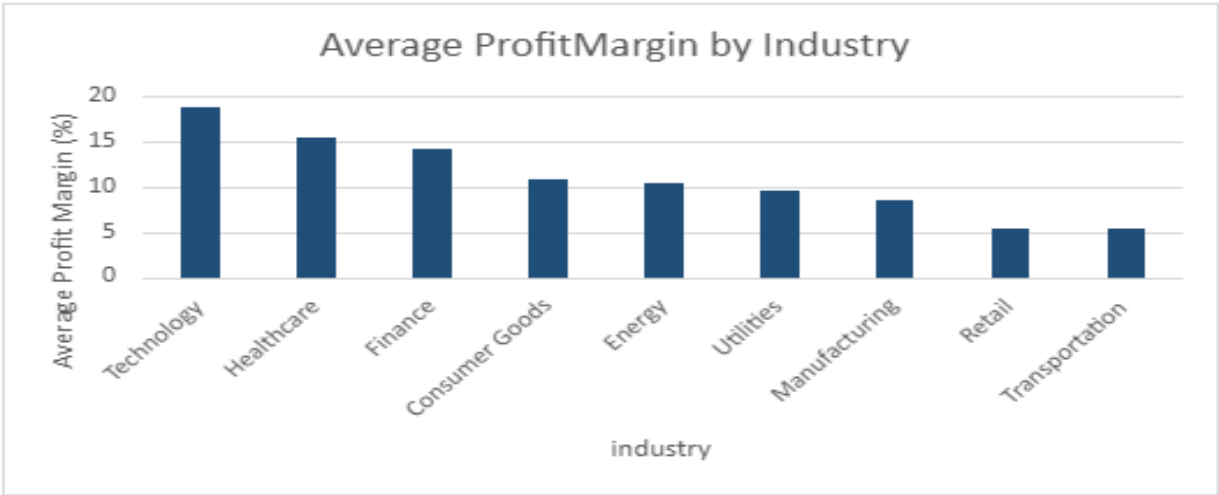
Key Findings and Visualisations

Which industry has the highest average ESG score by Industry?



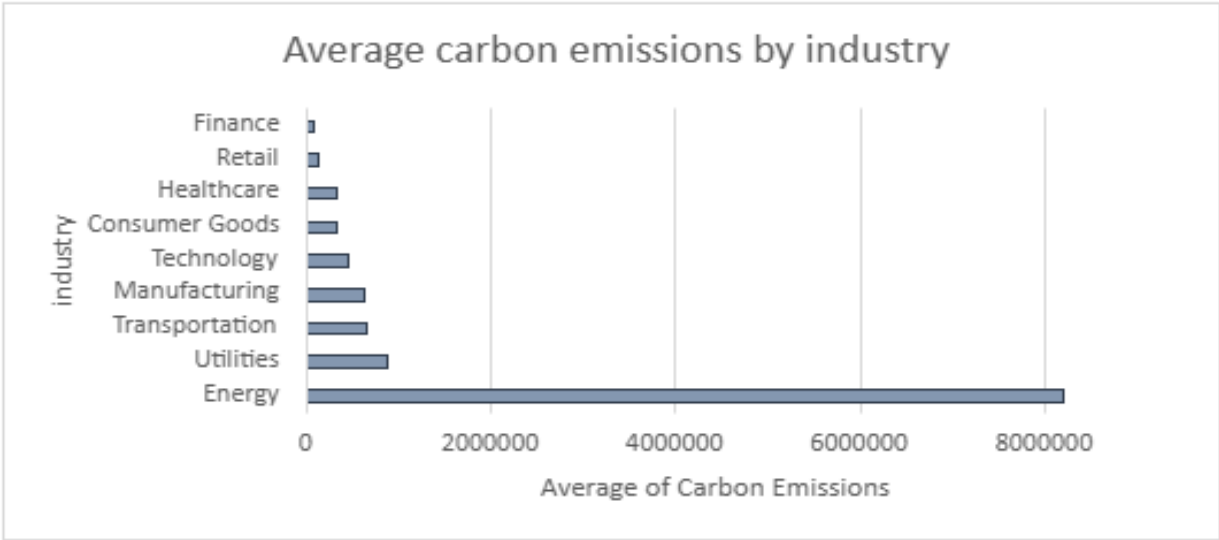
Finance and Technology have the highest average ESG score, with Finance at 64.6, while Transportation has the lowest with a 46.0 score.

Which Industry has the highest average profit margin?



Technology had the highest average profit margin at 18.8%, while transportation and retail had the lowest, at 5.3% and 5.5%

Which industry has the most carbon emissions?



The energy industry has by far the highest average carbon emissions in the dataset, with 8.19 million, while Finance has the lowest at only 7.2 thousand.

Business Insights

The results for which industries have the highest average ESG score showed that the industry with the highest ESG score is the finance sector, and the lowest ESG score is in transportation. However, this may be because industries might have different characteristics.

I decided to look at ESG individually to see if there are differences between industries' environmental, social, and governance performance.

The average ESG score for governance in the finance industry was the lowest, while consumer goods had the highest. This shows that finance doesn't have the strongest ESG performance. Although, the finance industry had the highest ESG score for environment, while transportation had the lowest. Industries like healthcare and transportation also score quite low for ESG social compared to other industries. This shows that ESG performance varies considerably across industries.

Across the 1000 companies, there is not a strong relationship between average ESG score and average profit margin. Some industries have higher average profit margins and have a high ESG score; the company-level correlation between ESG performance and profit margin is only 0.097, meaning there is little to no linear relationship between the two variables in this dataset.

The results illustrated that the Energy industry had the highest average carbon emissions, while finance had the lowest. This may be because raw emissions depend on company size and industry type, as a larger company may produce higher raw emissions due to the scale of operations. However, this cannot fully determine the actual measure of environmental performance.

When comparing company size with ESG performance, both larger and smaller companies showed a wide range of average ESG scores. This suggests that company size alone does not appear to have a clear relationship with ESG performance in this dataset. However, this was based on comparing the largest and smallest companies rather than calculating a formal correlation.

I compared companies' average ESG scores with their average profit margins to identify unusual combinations of ESG performance and profitability. I found several companies with relatively high ESG scores but low or negative profit margins, as well as companies with relatively lower ESG scores and high profit margins.

These examples show that ESG performance and profitability do not necessarily move together. This supports the correlation analysis, which found a correlation of 0.097, suggesting little to no linear relationship between the two variables in this dataset.

Limitations

- The dataset contained simulated company data, so the findings may not represent real-world companies.
- The correlation doesn't prove causation, so the analysis cannot determine whether ESG performance causes changes in profitability.
- Raw carbon emissions can be influenced by company size and industry, so they should not be used alone to assess environmental performance.
- The findings cannot be assumed to apply to all companies and industries, as ESG performance and profitability can vary across markets, countries and industries.

Conclusion

I analysed an ESG and financial performance dataset to investigate whether there was a relationship between ESG performance and financial performance. My analysis found that there was little to no linear relationship between average ESG performance and average profit margin, with a correlation of 0.097.

I was particularly interested in how company size and industry characteristics could affect the results. For example, raw carbon emissions varied significantly between industries, showing that company size and the nature of an industry need to be considered when interpreting ESG performance. Based on this analysis, I would recommend that businesses consider ESG performance alongside financial performance rather than assuming that stronger ESG Performance automatically leads to higher profits. The results showed that companies can have relatively high ESG scores alongside low profit margins, or lower ESG scores alongside higher profit margins.